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Extramedullary Myeloma whole genome sequencing reveals novel mutations in Cereblon, proteasome subunit G2 and the glucocorticoid receptor in multi drug resistant disease

Jan B. Egan[1,*], **K. Martin Kortuem**[1,*], **Ahmet Kurdoglu**[2], **Tyler Izatt**[2], **Jessica Aldrich**[2], **Rebecca Reiman**[2], **Lori Phillips**[2], **Angela Baker**[2], **Chang-Xin Shi**[1], **Jessica Schmidt**[1], **Winnie S. Liang**[2], **David W. Craig**[2], **John D. Carpten**[2], and **A. Keith Stewart**[1]

[1]Mayo Clinic, 13400 E. Shea Blvd, Scottsdale, Arizona 85259

[2]Translational Genomics Research Institute, 445 N. Fifth Street, Phoenix, Arizona 85004

Keywords

myeloma; *CRBN*; *NR3C1*; *PSMG2*; drug resistance

Extramedullary disease (EMD) in Multiple Myeloma (MM) is characterized by the detection of monoclonal plasma cells outside the bone marrow niche, and is frequently associated with poor prognosis.

Here we describe novel genomic events leading to drug refractory disease in a heavily pretreated 37-year-old IgG-kappa MM patient presenting with progressive, multi-drug refractory EMD. For the first time we report an acquired truncating mutation of Cereblon (*CRBN*) as well as point mutations in proteasome subunit G2 and the glucocorticoid receptor as an explanation for drug resistance. Initial myeloma treatment for the patient occurred over multiple years and included the immunomodulatory drugs (IMiDs) thalidomide and lenalidomide, the proteasome inhibitor bortezomib, cortiosteroids, radiation, one autologous and two allogeneic transplantations. She experienced extramedullary relapse, presenting as an extensive neck mass and smaller soft tissue nodules in the upper left triceps. The most recent therapy immediately prior to genomic sequencing was hyper-CVAD (hyperfractionated cyclophosphamide, vincristine, doxorubicin and dexamethasone) incorporating alkylating agent cyclophosphamide with transient minor response. The patient was then enrolled in a pilot study utilizing next generation sequencing (NGS) to identify novel markers and potential therapeutic targets. Samples were acquired with patient consent in compliance with Mayo Clinic Institutional Review Board. For this study we completed array comparative genomic hybridization, whole exome, whole genome (insert = 1 kb) and RNA sequencing (RNASeq) of a biopsy taken from the neck mass to thoroughly interrogate the tumour genome of this patient. The presence of mutations of

Corresponding Author: A. Keith Stewart, Mayo Clinic Collaborative Research Building, 13400 E Shea Blvd, Scottsdale, AZ 85259-5494, Phone: +1 480-301-4411, Fax: +1 480-301-8387, stewart.keith@mayo.edu.
*These authors contributed equally

Authorship contributions
JBE, KMK, WL, DWC, JDC and AKS contributed to experimental design, analysis and manuscript preparation. AK, TI, JA conducted analysis. WL, RR, LR and AB conducted sequencing. KMK, CXS and JS conducted validation studies.

Conflicts of interest
There are no conflicts of interest to declare.

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interest was evaluated by capillary sequencing in an expanded cohort of 25 CD138+ MM samples with low *CRBN* expression.

The neck mass pathology confirmed sheets of atypical plasma cells, kappa light chain restriction, CD138+, CD20− and CD45−. Array comparative genomic hybridization revealed multiple copy number abnormalities, most notably del1(p13.2–34.2), monosomy 13 and monosomy X. Therapy subsequent to biopsy for genome sequencing was with pomalidomide and dexamethasone without response. Unfortunately, the patient succumbed to her disease in less than the 12 weeks required at the time for sequencing and data analysis.

Sequencing revealed a highly disturbed genome (Figure 1) consisting of 4 somatic insertions/deletions, 38 intra-chromosomal rearrangements, and 35 translocations, including the high risk marker and initiating tumour event t(14;16). Furthermore, 271 nonsynonymous, somatic point mutations were detected in genes including *KRAS*, *PIK3CA*, *ATM*, and *NFKB2* (Table I). Importantly, a Q99* truncating mutation as well as a R283K point mutation were observed in *CRBN*, that we recently demonstrated as essential for the anti-MM action of IMiDs (Zhu, *et al* 2011). To our knowledge this is the first documented mutation of Cereblon in a primary myeloma sample. Additional sequencing of *CRBN* in the expanded cohort of 25 patients revealed a synonymous mutation in only one sample.

We also observed in the patient biopsy a potentially clinically relevant nonsynonymous point mutation in proteasome assembly chaperone 2, *PSMG2* (E171K). PSMG2 is a proteasome assembly protein involved in mammalian 20S proteasome maturation (Hirano, *et al* 2005). Mutations in proteasome assembly components contribute to proteasome inhibitor resistance (Keats, *et al* 2007), possibly explaining this patient's bortezomib-refractory disease. Capillary sequencing of *PSMG2* in our expanded cohort revealed no mutations, although exonic deletion of *PSMG2* has also been reported in myeloma (Walker, *et al* 2012).

The last nonsynonymous point mutation associated with drug resistance was identified in *NR3C1* (G369A), a glucocorticoid receptor. Mutation of *NR3C1* has been associated with resistance to steroid therapy (Bray and Cotton 2003), which this patient received and proved refractory. No *NR3C1* mutations were identified in our expanded cohort and none have been previously reported in other myeloma genomes (Chapman, *et al* 2011, Walker, *et al* 2012). Mutations in *NR3C1* have however been described in the glucocorticoid resistant MM.1R cell line (Moalli, *et al* 1992). Patients with low *NR3C1* expression levels who received thalidomide demonstrated better progression-free survival and overall survival than those with low *NR3C1* who did not receive thalidomide (Heuck, *et al* 2012).

While these mutations suggest causality of drug-refractory disease, they do not identify pathways that can be exploited with targeted therapies. Additional mutations were observed in pathways for which targeted therapies are available. This patient had mutations in *KRAS* (G12C) and in *ATM* (T1985I), both of which affect the signalling of MEK downstream, thus making MEK a therapeutic target of interest in this patient. While there are no approved MEK inhibitors available for MM treatment, more than 100 trials are currently investigating MEK inhibitors, of which three of these trials are being conducted in MM patients (www.clinicaltrials.gov).

The patient tumour also contained canonical, activating mutations in *PIK3CA* (E542K). Interestingly, one study demonstrated that 64% of *PIK3CA* mutations occur in exon 9, where codon 542 is located. Moreover, 19% of patients with *PIK3CA* mutations also presented with *KRAS* mutations, of which 9% are G12C (Janku, *et al* 2012), found in our patient. The PI3K pathway is vitally important as it regulates downstream targets, such as

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AKT and MTOR, which are responsible for cell proliferation, growth, survival and metastasis (Bartholomeusz and Gonzalez-Angulo 2012). In addition, a number of clinical trials are currently investigating PIK3 inhibitors (www.clinicaltrials.gov).

In summary, this is the first description of *CRBN* mutations in a primary myeloma sample and furthermore of a "triple negative" MM patient possessing mutations probably contributing to resistance to all three major drug classes utilized in MM therapy. These mutations were not replicated in our validation cohort of 25 patients with low level *CRBN* expression and functional data have not yet been obtained, thus further investigation is necessary to better understand the mutation frequency and the functional significance of mutation in these genes. In summary, our approach utilizing comprehensive next generation sequencing not only identified mutations suggestive of the patient's refractory disease, but also revealed unforeseen therapeutic options highlighting the importance of this technology in advancing individualized medicine.

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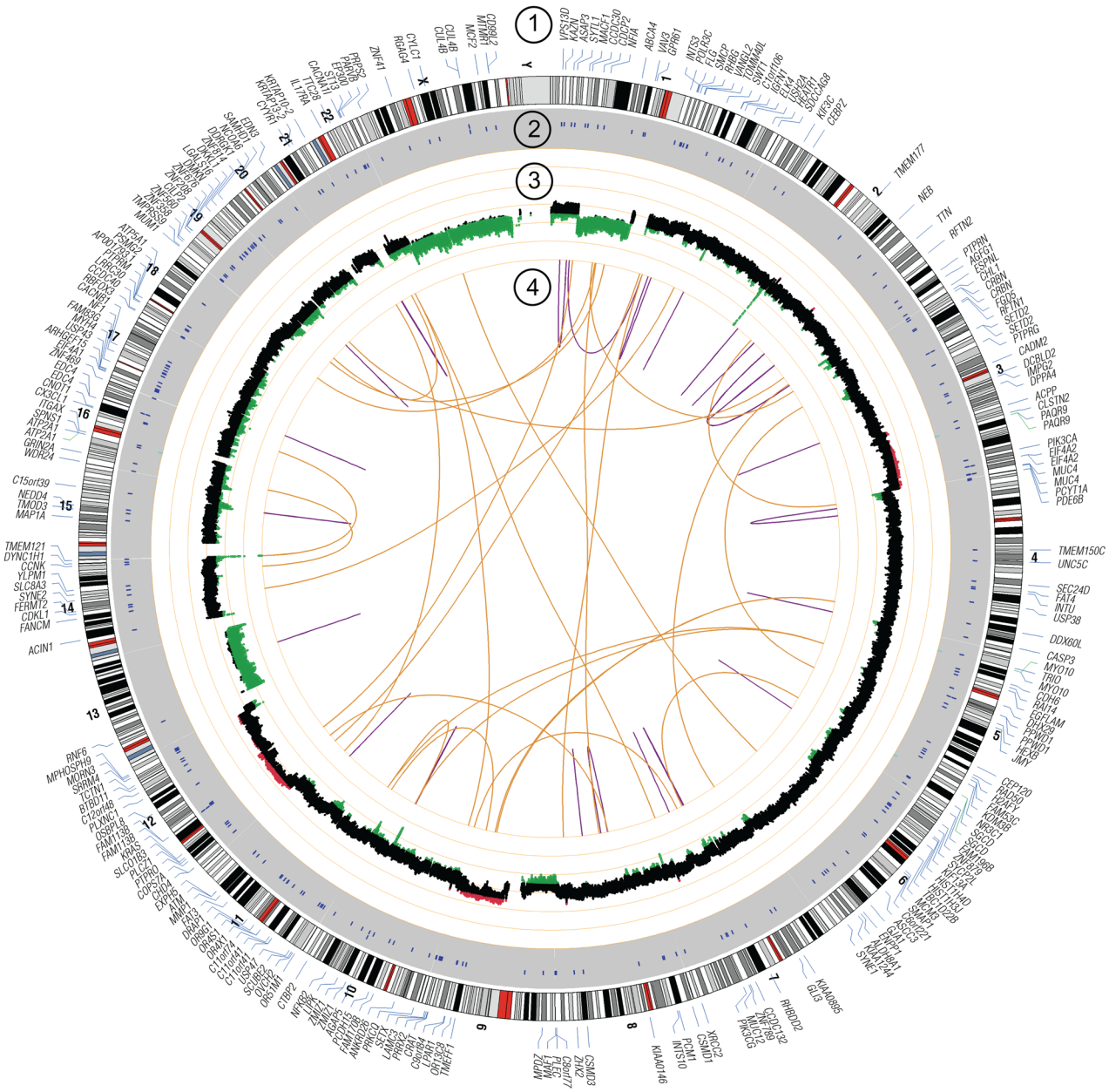

Figure 1.

Circos plot depicting genome wide somatic variants, rearrangements and copy number changes derived from next generation sequencing. Numbers with circles around them indicate the following: 1) somatic single nucleotide variation (SNV), 2) location of SNV and insertion/deletions, 3) copy number gain (red) and copy number loss (green), and 4) translocations (orange) and intrachromosomal rearrangements (purple).

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Table I

Summary of clinically relevant single nucleotide variations

Chr	hg19 position	SNV	Gene	SIFT	Polyphen2	Effect	Amino acid	dbSNP
3	3195747	C>T	*CRBN*	Tolerated	Benign	Nonsynonymous coding	R283K	
3	3215822	G>A	*CRBN*	Not predicted	Not predicted	STOP gained	Q99*	
3	178936082	G>A	*PIK3CA*	Tolerated	Damaging	Nonsynonymous coding	E542K	rs121913273
5	142779299	C>G	*NR3C1*	Tolerated	Damaging	Nonsynonymous coding	G369A	
10	104160420	C>T	*NFKB2*	Damaging	Damaging	Nonsynonymous coding	P603S	
11	108183173	C>T	*ATM*	Tolerated	Damaging	Nonsynonymous coding	T1985I	
12	25398285	C>A	*KRAS*	Damaging	Damaging	Nonsynonymous coding	G12C	rs121913530
17	29588752	G>A	*NF1*	Tolerated	Damaging	Nonsynonymous coding	R1534Q	
18	12720612	G>A	*PSMG2*	Tolerated	Benign	Nonsynonymous coding	E171K	

Chr, chromosome; SNV, single nucleotide variation; SIFT, Sorting Intolerant From Tolerant program; Polyphen2, Polymorphism Phenotyping v2 tool; dbSNP, Single Nucleotide Polymorphism Database